

BEATS: Audio Pre-Training with Acoustic Tokenizers

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- Audio powers search, safety, accessibility, and ambient intelligence.
- SSL enables learning from massive unlabeled audio across domains.
- Unified audio representations unlock transfer to many downstream tasks.

Why General Audio Is Hard

- Audio is diverse: speech, music, events, overlapping sources, noise.
- Existing SSL often reconstructs low-level spectrogram details.
- We need semantics-focused pre-training robust to nuisance factors.

Limitations of Reconstruction-Based Audio SSL

- Optimizes pixel/TF fidelity, not event semantics.
- Sensitive to reverberation, channel, and noise perturbations.
- Inefficient: reconstructs masked content at high resolution.
- Lags behind discrete prediction paradigms in vision/speech/NLP.

- Replace reconstruction with masked discrete label prediction for audio.
- Iteratively co-train an acoustic tokenizer and an audio SSL model.
- Design tokenizers that produce semantic, noise-robust labels.
- Efficient ViT-based Masked Audio Modeling with 75% masking.
- SOTA on AudioSet and ESC-50 with fewer parameters, no external data.

- Tokenizer labels audio patches with discrete codes.
- SSL model predicts masked labels from unmasked context.
- Trained SSL model distills a better tokenizer; repeat.

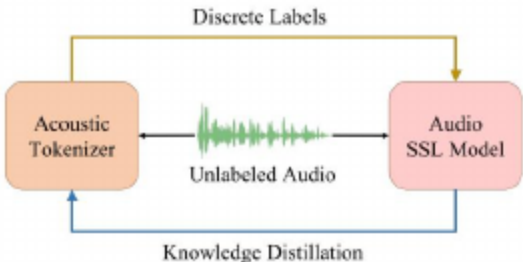


Figure: Iterative audio pre-training of BEATS.

- Random-Projection Tokenizer: cold-start labels via nearest codebook.
- Self-Distilled Tokenizer: Transformer + VQ with learnable codebook.
- Distillation from SSL or supervised teacher injects semantics.

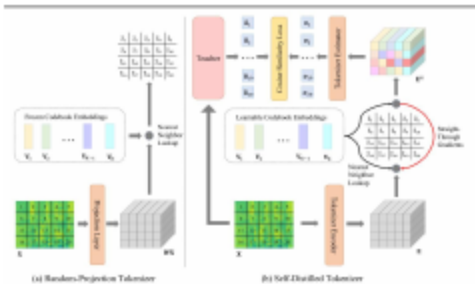


Figure: Acoustic tokenizers for discrete label generation.

- Random linear projection maps patches to embedding space.
- Fixed codebook assigns nearest index as discrete label.
- Simple, fast, and sufficient to bootstrap BEATSiter1.

- Transformer encoder produces patch embeddings.
- Vector Quantization assigns codes; EMA-updated codebook.
- Estimator learns to match teacher outputs (cosine loss).
- Teacher can be SSL-pretrained or supervised fine-tuned.

- Mask 75% of patches; encode only unmasked for efficiency.
- Predict tokenizer labels at masked positions (cross-entropy).
- ViT backbone with relative positions and DeepNorm.

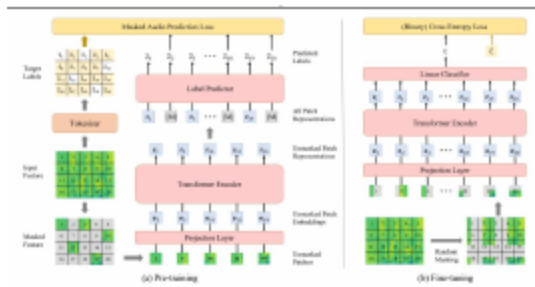


Figure: Overview of audio SSL model pre-training and fine-tuning.

- Feed only unmasked patches into the encoder.
- Heavy masking boosts speed and regularization.
- Discrete targets simplify the prediction head.

- Discard label predictor; keep encoder as backbone.
- Mean-pool features; add linear head (CE/BCE).
- Use SpecAug and mixup where appropriate.

- Alternate improving labels (tokenizer) and model (SSL).
- Formulated as EM-like optimization over latent labels.
- Guarantees non-decreasing data likelihood per iteration.

- Pre-train on AudioSet-2M; evaluate on AS-2M/20K (mAP).
- Transfer to ESC-50, Speech Commands (V1/V2), IEMOCAP ER (acc).
- Model: ViT Base (90M) with relative positions and DeepNorm.

- Compare to strong supervised and SSL baselines (e.g., ViT-based and MAE-style).
- No external labeled data for BEATS; fair transfer protocols.
- Strong augmentation and class balancing per task when needed.

- AS-2M mAP 48.6 with a Base ViT, no external data
- AS-20K mAP 38.9
- ESC-50 accuracy 98.1%
- BEATSiter1 already beats reconstruction SSL; self-distilled tokenizers add gains

- Tokens emphasize stable cues over nuisance variations.
- Iterative distillation improves low-resource performance.
- Unified objective simplifies cross-task transfer pipelines.

- Iteration adds compute overhead (tokenizer + model loops).
- Reliance on AudioSet; broader corpora may help coverage.
- Model size is modest vs frontier; scaling likely beneficial.

Acknowledgments

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Open Questions

- How to scale tokenizers without labels?
- Can tokens unify cross-domain audio tasks?
- How to co-train audio with vision/language tokens?
- What guarantees beyond EM-style analysis?
- How do tokens evolve with model/data scale?